Chapter 1

Introduction to Text

In computer systems, text is represented in files as a sequence of characters, each of which corresponds to a specific number, known as a character code. These character codes are then stored in the file as binary data.

1.1 Newlines and Carriage Returns

Two of the character codes that have special meanings are the newline (often represented as '\n') and the carriage return (often represented as '\r').

The newline character signifies the end of a line of text and the beginning of a new one. If you parse a text file with a Python script, it will see your 'enter' key press as a new line symbol. The carriage return character moves the cursor to the beginning of the line. The use of these characters can vary between operating systems. Unix-based systems (like Linux and MacOS) use the newline character to indicate the end of a line, while Windows systems use a combination of a carriage return and a newline ('\r\n').

1.2 ASCII

The American Standard Code for Information Interchange (ASCII) is one of the earliest character encodings. It uses 7 bits to represent each character, allowing it to define up to $2^7 = 128$ different characters. These include the English alphabet (in both lower and upper cases), digits, punctuation symbols, control characters (like newline and carriage return), and some other symbols.

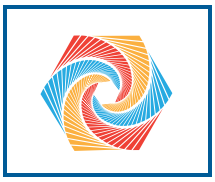
1.3 UTF-8

UTF-8 (8-bit Unicode Transformation Format) is a variable-width character encoding that can represent every character in the Unicode standard, yet remains backward-compatible with ASCII. For the ASCII range (0-127), UTF-8 is identical to ASCII. However, it can use additional bytes (up to 4 bytes in total) to represent characters that are not included in ASCII, such as characters from other languages, emojis, and many other symbols. This has made UTF-8 a widely used encoding in many modern systems.

This is a draft chapter from the Kontinua Project. Please see our website (https://kontinua.org/) for more details.

Appendix A

Answers to Exercises



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